MIL-B-46957B (PA)
5 January 1977
SUPERSEDING
MIL-B-46957A (MU)
15 April 1964

MILITARY SPECIFICATION
BURSTER, PROJECTILE, M48
LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by Picatinny Arsenal Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of burster designated as Burster, Projectile, M48.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-48078 - Ammunition, Standard Quality Assurance Provision, General Specification for

STANDARDS

MILITARY

MIL-SDT-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-453 - Inspection, Radiographic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1315

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

8796522 - Marking Diagram and Sealing for Wooden Packing Boxes.
8864750 - Box, Packing, Ammunition, for Projectile Burster, M48 for Projectile, 105MM, WP-T, M416
8864751 - Carton, Packing Ammunition for Projectile, Burster, M48 for Projectile, 105MM, WP-T, M416
8886484 - Burster, Projectile, M48

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Burster.-The burster shall comply with all requirements specified on Drawing (dwg.) 8886484 and with all requirements specified in applicable specifications.

3.3 Cavitation

3.3.1 Area from bottom to one inch above bottom of tube.-There shall be no cavities, porous areas, cracks or annular rings in the charge in this area.

3.3.2 Area from top to one inch from bottom of tube.-There shall be no single cavity having a projected length of more than one inch or a sum total of cavities of more than 3/8 square inch and there shall be no more than three transverse cracks 1/32 inch wide, maximum, in this area.

3.3.3 Separation of charge from base of tube.-There shall be no void greater than 1/32 inch between the charge and the base of the tube.

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3.3.4 Porosity.-Porous areas shall be treated as cavities, and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.4 Workmanship.-All parts shall be free from chips, dirt, cracks, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Quality Conformance Inspection

4.2.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of Burster, Projectile M48 shall contain:

a. Composition B4 from not more than one lot.
b. Composition B from not more than one lot.
c. Metal parts from one manufacturer from not more than one lot interfix number.

4.2.2 Examination.-See MIL-A-48078

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105 or MIL-STD-1235.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-46957B (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.1	Tube, burster, prior to loading	1 OF 1	8886483
			NEXT HIGHER ASSEMBLY 8886484

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Foreign matter in interior.		0.40%	3.2	Visual
Minor:					
201	Total length, minimum (min.)		0.65%	3.2	Gage
202	Thickness through bottom, min.		0.65%	3.2	Gage
203	Outside diameter, min.		0.65%	3.2	Gage
204	Finish improper		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-46957B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.2	Projectile Burster				8886484
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Cushioning pad missing or insecure			3.2	Visual
2.	Burster tube cracked			3.2	Visual
3.	Cavitation	4.2.3.1		3.3	4.3.1
Major:					
101	Failure to chamber gage		0.40%	3.2	Gage
102	Total length, maximum (without compression of cushioning pad)		0.40%	3.2	Gage
103	Cavitation	4.2.3.1		3.3	4.3.1
Minor:					
201	Cushioning pad not centered		0.65%	3.2	Visual
202	Marking misleading or unidentifiable including incorrect color code and arrow.		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-46957B (PA)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.2.2.3	Carton, Prior to sealing	1	1	9964751
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major: 101	Burster projectile missing		0.40%	3.2	Visual
Minor:	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-46957B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.4	Carton, sealed				8864751
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Tape improperly applied		0.40%	3.2	Visual
Minor:					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Contents loose upon shaking		0.65%	3.2	Manual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-46957B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8864751
4.2.2.5	Carton in bag, sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Bag improperly sealed torn or cut	4.2.3.2	0.40%	3.2	Visual
102	Heat seal			3.2	
Minor:	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-46957B (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.6	Sealed wood packing box	1 OF 1	8864750
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Box damaged		0.40%	3.2	Visual
102	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
Minor:					
201	Strapping improperly assembled		0.65%	3.2	Visual
202	Lid improperly assembled		0.65%	3.2	Visual
203	Marking misleading or unidentifiable		0.65%	3.2	Visual
204	Contents loose upon shaking		0.65%	3.2	Manual

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4.2.3 Testing

4.2.3.1 Cavitation.-Cavitation defects are classified as follows:

Critical

Any cavity, crack or annular ring in the area from the bottom of tube to one inch above the bottom (see 3.3.1)

Any separation of charge from the base of the tube in excess of 1/32 inch (see 3.3.3)

Major

Cavitation or cracks in excess of that permitted by 3.3.2 and 3.3.4.

The sampling plan and procedure for this test for critical and major defects shall be in accordance with MIL-STD-1235. Using a CSP 1 plan, the AQL for critical defects shall be .010 percent and for major defects 0.25 percent. The appropriate production interval shall be specified by the procuring activity.

4.2.3.2 Heat seal of sealed carton (see dwg. 8864751).-Sampling acceptance criteria and test procedures for this test shall be as prescribed in Specification MIL-P-116 for the "Heat Seal Test".

4.2.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.2.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.3 Test methods and procedures

4.3.1 Determination of cavities.-The sample bursters shall be subjected to x-ray examination. In all exposures the x-ray beam shall form a 90 degree angle with the longitudinal axis of the burster. Examination of the x-ray negative shall be made to determine compliance with applicable requirements.

Note.-Radiographic equipment, operations and procedures shall be qualified in accordance with Standard MIL-STD-453. In addition, the sensitivity of x-ray technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.3.1.1 and 4.3.1.2. In the event an original x-ray picture leaves doubt as to acceptability, the burster shall be re-x-rayed at an angle 90 degrees from that of the original exposure and in the same plane. Acceptability shall be determined on the basis of the film giving the largest projected area of defect.

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4.3.1.1 Check of sensitivity of x-ray technique.-The sensitivity of x-ray technique shall be determined by means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the burster away from the film (source side). Observation shall be made of the image of the penetrameter on the x-ray film. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter, including the hole in the center, is sharply defined on the x-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thickness and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole, the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see Standard MIL-STD-453. The actual thickness may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated in Figure 1 of this specification.

4.3.1.2 Check of photographic density of film.-The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. To judge densities it is recommended that standards be used (6.3). Comparison of the film with standards may be made visually, however, the use of a densitometer is recommended.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The M48 projectile burster shall be preserved and packaged in accordance with dwg. 8864751.

5.2 Packing

5.2.1 Level A.-The cartons shall be packed in accordance with dwg. 8864750.

5.3 Marking.-Marking shall be in accordance with dwgs. 8864751 and 8864750 and 8796522 when applicable.

6. NOTES

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6.1 Ordering data.-See MIL-A-48078.

6.2 Submission of design for Approval.-See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T, New Jersey, 07801.

6.3 Film density.-"American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, N.Y., has been found satisfactory as a source of information on film densities.

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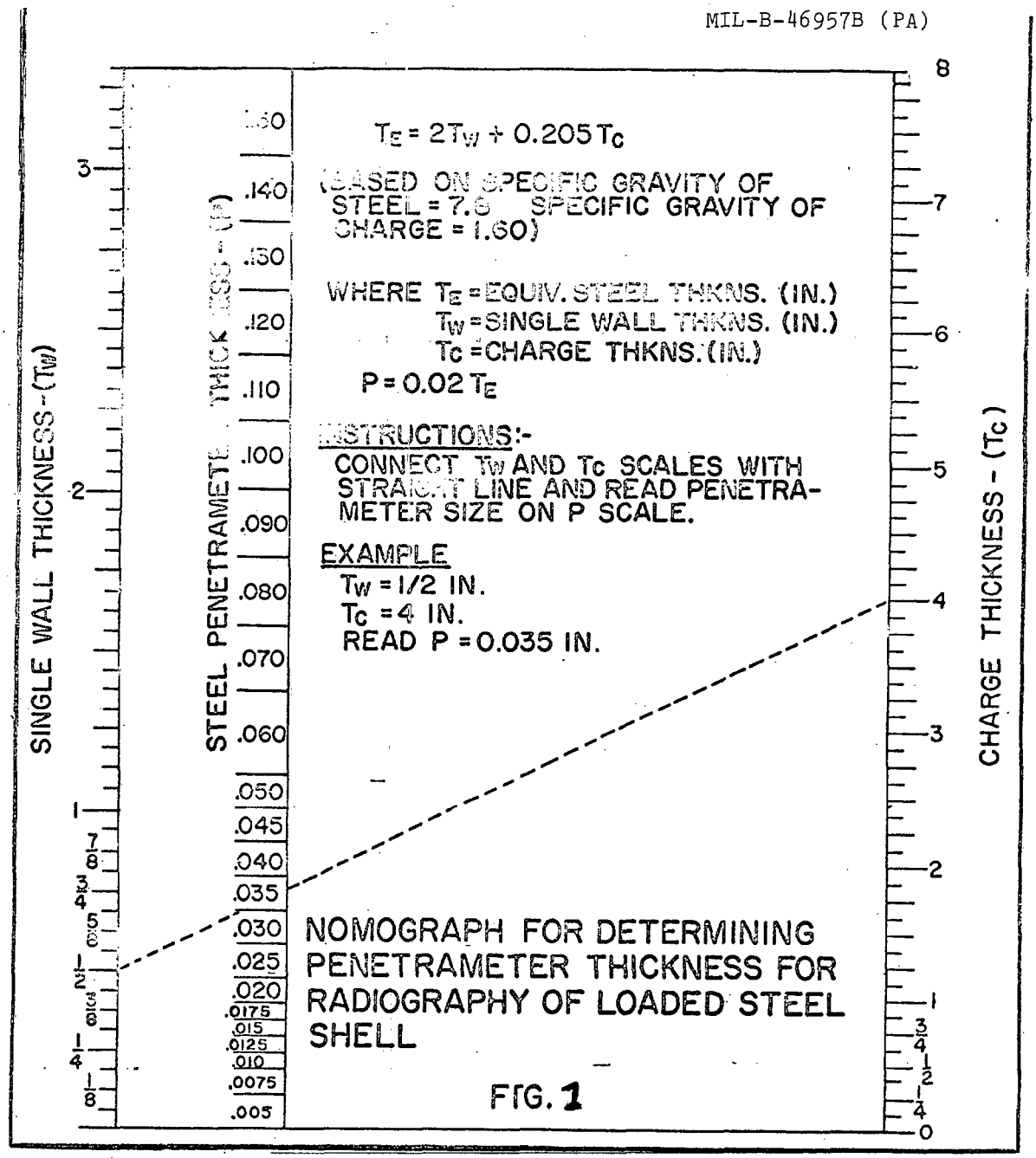


FIG. 1

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NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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